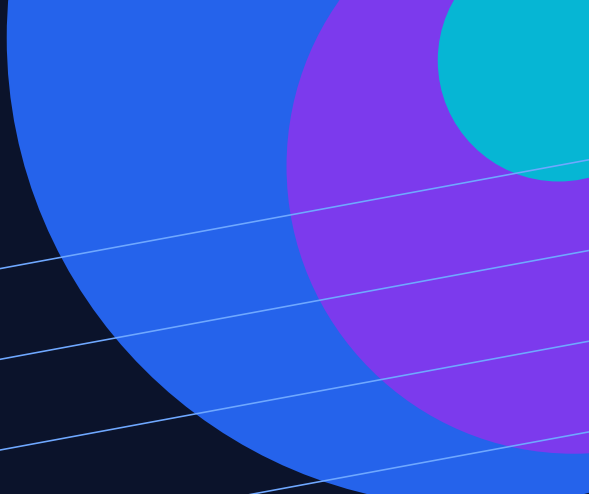




THREAT RESEARCH / DETECTION ENGINEERING

Far-Future Calendar Access

Generalized HOLLOWGRAPH discovery experiment

REPORT	STATUS	CONFIDENCE
THREAT HUNTING STUDY	HUNTING QUERY	HIGH OBSERVABILITY / MEDIUM MALICIOUSNESS

Prepared for defensive research and tenant-specific validation. This report separates source-observed facts from analytical interpretation and proposed detection logic.

VERSION 1.0 / 22 JULY 2026

Threat Overview

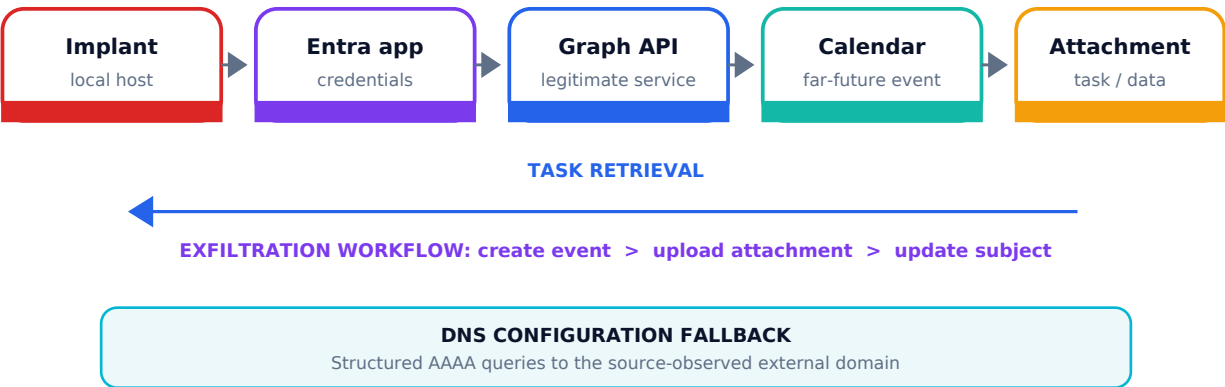
WHY THIS MATTERS

HOLLOWGRAPH demonstrates how a compromised Microsoft 365 identity and a legitimate SaaS API may be repurposed as a covert two-way channel. Destination-based controls may see ordinary Microsoft Graph traffic while the operational meaning is hidden in calendar objects.

Source-observed behavior

Stage	Observed fact	Defensive relevance
Configuration	The analyzed .NET implant used Entra ID application credentials stored locally in logAzure.txt.	A local credential artifact and application identity provide endpoint and identity investigation pivots.
Task retrieval	The implant issued GET requests to /users/{mailbox}/calendarView for 13 May 2050 and read tasking from an attached file.	The endpoint and far-future date are present in the request URI and may be observable in Graph Activity Logs.
Exfiltration	The implant created an event, uploaded encrypted data as attachments, and changed the event subject.	The sequence is source-supported, but body content needed for specificity is not documented in Graph Activity Logs.
Credential refresh	Structured AAAA queries used a source-observed external domain to update configuration values.	Useful for IOC-led hunting and DNS investigation; insufficient alone for a durable generic rule.

Attack-channel model



Evidence Boundaries

Evidence class	Statement	Use in this package
Source fact	The analyzed sample queried calendarView with a date range on 2050-05-13.	Directly supports the exact-date production candidate.
Documented telemetry	MicrosoftGraphActivityLogs documents request method, URI, response status and identity/application context.	Supports filters and investigation fields without assuming body visibility.
Analyst interpretation	Far-future calendar access may be anomalous and may generalize beyond the exact sample.	Supports hunting only; maliciousness is not established.
Unvalidated assumption	The tenant preserves startDateTime in RequestUri without truncation or transformation.	Must be tested before deployment or promotion.

REQUEST METADATA

method, URI, status

VISIBLE

IDENTITY CONTEXT

app, SP, user, IP

VISIBLE

REQUEST BODY

subject, event date

NOT DOCUMENTED

ATTACHMENT BODY

name and content

NOT DOCUMENTED

SAFE DETECTION BOUNDARY

Use RequestMethod + RequestUri + ResponseStatusCode and documented identity fields.

EXCLUDED FROM THE RULE

Event subject, attachment filename, attachment type and payload content.

CORE CONTROL

No query in this package infers calendar subject, attachment filename, content type, event body or payload content. Those values are excluded unless a separate, validated telemetry source exposes them.

Hunting Design

HUNTING ONLY

BASELINE REQUIRED

2040-2099

Hunting hypothesis

HYPOTHESIS	Applications querying calendarView for years 2040 through 2099 are likely uncommon. Reviewing that activity may reveal HOLLOWGRAPH-like dead-drop access even when the attacker changes the exact source-observed date.
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Condition	Analytical purpose	Caution
GET	Focus on calendar retrieval activity.	Does not prove task retrieval.
/calendarView	Anchor the hunt to the observed endpoint.	Legitimate applications use this endpoint.
startDateTime=20[4-9][0-9]-	Generalize to far-future years while remaining reviewable.	Range is analyst-selected, not source-validated as a universal threshold.
No status filter	Retain failed probes for context.	Increases benign and malformed traffic.
No aggregation	Avoid an unsupported count threshold.	Analysts must group results during review.

KQL hunting query

```
MicrosoftGraphActivityLogs
| where TimeGenerated >= ago(30d)
| where RequestMethod == "GET"
| where RequestUri contains "/calendarView"
| where RequestUri matches regex @"(?:)[?&]startDateTime=20[4-9][0-9]-"
| project TimeGenerated, AppId, ServicePrincipalId, UserId,
        IPAddress, UserAgent, RequestUri, ResponseStatusCode
```

Hunt Workflow

1

DISCOVER

Run 30 days
then retention

2

CLUSTER

Group by app,
SP, user, UA

3

ATTRIBUTE

Find owner
and purpose

4

ESCALATE

Investigate
unknowns

Step	Action	Expected output
1. Discovery	Run the query for 30 days; extend to full retention after confirming performance.	Raw population of far-future calendarView requests.
2. Clustering	Summarize offline by AppId, ServicePrincipalId, UserId, UserAgent, IP and status.	Stable recurring patterns versus isolated activity.
3. Attribution	Map each application to owner, permissions, business purpose and expected mailbox scope.	Approved, unexplained or suspicious classification.
4. Context expansion	Review app sign-ins, credential changes, consent, endpoint activity and targeted mailbox context.	Evidence supporting or weakening compromise hypothesis.
5. Documentation	Record URI format, false positives and identity-based exclusions.	Promotion evidence or explicit rejection rationale.

**DO NOT
AUTO-ALERT**

The selected 2040-2099 range is a practical experiment, not a proven malicious threshold. The hunt should remain analyst-driven until tenant history demonstrates stable specificity.

Telemetry & Analytical Pivots

Field	Hunt use	Validation question
RequestUri	Extract endpoint and future year condition.	Are query parameters retained and readable?
RequestMethod	Restrict to retrieval behavior.	Is casing consistent?
ResponseStatusCode	Separate accepted, denied and malformed attempts during analysis.	Is it populated consistently?
AppId / ServicePrincipalId	Primary application clustering and attribution.	Can identities be resolved to inventory and owners?
UserId	Identify mailbox/user context where populated.	What does null mean for application-only activity?
IPAddress / UserAgent	Compare source and client patterns.	Are values stable, proxied or shared?

Triage priority model

Priority	Characteristics	Recommended response
High	Unknown application; privileged Graph roles; isolated far-future access; unusual source context.	Immediate identity and endpoint investigation.
Medium	Known application but unexpected mailbox, timing, URI range or source.	Confirm owner and inspect change/sign-in history.
Low	Approved test or migration tool with documented recurring behavior.	Record evidence; consider narrow identity-based tuning.

False Positives & Promotion Criteria

Scenario	Why it may match	Evidence that reduces concern
QA / integration testing	Extreme dates are common boundary tests.	Approved test owner, expected environment and change record.
Migration / archival tooling	Products may query broad calendar ranges.	Documented product behavior and stable approved Appld.
Long-range planning	Business software may model future schedules.	Expected mailbox scope and recurring predictable pattern.
Malformed internal software	Bad date generation may produce future years.	Known defect, owner confirmation and failed statuses.

Promotion gates

Requirement	Promote only when...
URI fidelity	startDateTime is consistently visible across validated request variants.
Low legitimate volume	Historical matches are rare, understood and attributable.
Stable exclusions	Approved applications can be excluded using application identity and documented scope.
Reproducibility	Controlled far-future test traffic is detected across expected encodings.
Investigation context	Application, service principal, user, IP, user agent and response context are available.

REJECTION CONDITIONS

Do not promote when RequestUri omits the parameter, URI encoding causes unreliable matching, normal applications create unresolved recurring results, or identity context is insufficient to distinguish approved activity.

ATT&CK; Mapping & References

Technique	Relationship to observed behavior	Mapping posture
T1102.002 - Web Service: Bidirectional Communication	Microsoft Graph and calendar objects are used as a two-way dead-drop channel.	Direct behavioral relationship.
T1071.001 - Application Layer Protocol: Web Protocols	The implant communicates with Microsoft Graph over web protocols.	Supporting protocol relationship.

Primary threat research

1. Group-IB. HOLLOWGRAPH: Turning Microsoft Graph into a Covert C2 Channel.
group-ib.com/blog/hollowgraph-microsoft-365/

Platform and schema documentation

2. Microsoft Learn. Microsoft Graph activity logs overview.
learn.microsoft.com/en-us/graph/microsoft-graph-activity-logs-overview
3. Microsoft Learn. MicrosoftGraphActivityLogs table reference.
learn.microsoft.com/en-us/azure/azure-monitor/reference/tables/microsoftgraphactivitylogs

MITRE ATT&CK;

4. MITRE ATT&CK; T1102.002 - Web Service: Bidirectional Communication.
attack.mitre.org/techniques/T1102/002/
5. MITRE ATT&CK; T1071.001 - Application Layer Protocol: Web Protocols.
attack.mitre.org/techniques/T1071/001/

ANALYTICAL POSTURE

High confidence that far-future calendarView access can be searched when RequestUri retains startDateTime. Medium confidence that a match represents malicious behavior. The query is deliberately classified as hunting until environmental evidence supports a production decision.

Document controls

Field	Value
Research date	22 July 2026
Version	1.0
Audience	SOC analysts, detection engineers, threat hunters
Data handling	No customer data or internal indicators included
Review trigger	New source evidence, schema change, tenant test results or material false-positive findings